GATE ME 90 Days Challenge

Day 49/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Applied Thermodynamics	**IC Engines**	**54 days to GATE**

How to use this set: 15 questions (8 MCQ, 7 NAT) on the air-standard Otto, Diesel and dual cycles (efficiency, mean effective pressure, cut-off and pressure ratios, comparisons) and IC engine performance: indicated and brake power, mechanical, thermal, volumetric and relative efficiencies, bsfc, the Morse test and the heat balance. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

The air-standard efficiency of the Otto cycle depends on

(A) the heat supplied per cycle

(B) the compression ratio and γ only

(C) the maximum cycle temperature

(D) the engine speed

Q2 NAT

An air-standard Otto cycle (γ = 1.4) has an efficiency of 55 %. Its compression ratio (round off to two decimal places) is _____

Q3 MCQ

In an air-standard Diesel cycle 1–2–3–4 (1–2 isentropic compression, 2–3 constant-pressure heat addition), the cut-off ratio is defined as

(A) V_1/V_2

(B) p_3/p_2

(C) V_3/V_2

(D) V_4/V_3

Q4 NAT

An air-standard Diesel cycle (γ = 1.4) has a compression ratio of 16 and a cut-off ratio of 2.5. Its thermal efficiency (in %, round off to two decimal places) is _____

Q5 MCQ

Although the Otto cycle is more efficient than the Diesel cycle for the same compression ratio, practical CI (diesel) engines are more efficient than SI engines mainly because

(A) diesel fuel has a much higher calorific value

(B) CI engines run at higher speeds

(C) CI engines have lower friction

(D) CI engines work at much higher compression ratios (about 15–22), as only air is compressed

Q6 NAT

An engine has an indicated power of 60 kW and a mechanical efficiency of 80 %. It burns 15 kg/h of fuel of calorific value 42 000 kJ/kg. Its brake thermal efficiency (in %, round off to two decimal places) is _____

Q7 MCQ

In a rope brake dynamometer, W is the dead load, S the spring balance reading, D the effective brake-drum diameter (m) and N the speed (rpm). The brake power (kW) is

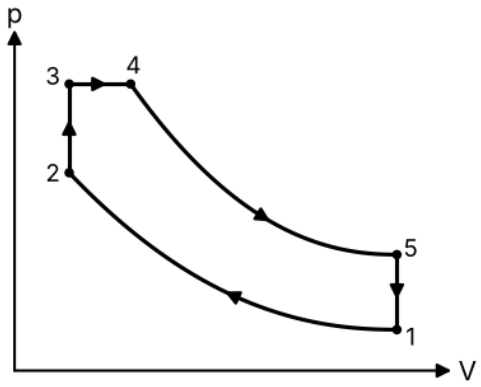
(A) (W − S)πDN/60 000
(B) (W + S)πDN/60 000
(C) (W − S)DN/60 000
(D) 2π(W − S)N/60 000

Q8 NAT

A four-cylinder four-stroke engine (bore 90 mm, stroke 100 mm) develops a brake power of 40 kW at 3000 rpm. Its brake mean effective pressure (in bar, round off to two decimal places) is _____

Q9 MCQ

The p–V diagram shown is that of an air-standard dual (limited-pressure) cycle. In this cycle heat is added



(A) during 2–3 at constant volume and during 3–4 at constant pressure
(B) during 2–3 at constant pressure and during 3–4 at constant volume
(C) only during 2–3 at constant volume
(D) during 4–5 at constant volume

Q10 NAT

An air-standard dual cycle (γ = 1.4) has a compression ratio of 15, a pressure ratio of 1.4 during constant-volume heating and a cut-off ratio of 1.5. Its thermal efficiency (in %, round off to two decimal places) is _____

Q11 MCQ

Beyond a certain speed, the volumetric efficiency of an engine falls as the speed increases, mainly because

(A) the compression ratio falls
(B) the flow losses (friction and choking) across the intake valve rise sharply with the gas velocity
(C) the fuel becomes richer
(D) the exhaust temperature falls

Q12 NAT

An air-standard Otto cycle has a compression ratio of 8. At the start of compression the air is at 100 kPa and 300 K, and 1500 kJ/kg of heat is added (c_v = 0.718 kJ/kg·K, γ = 1.4). The maximum pressure in the cycle (in MPa, round off to two decimal places) is _____

Q13 MCQ

The relative efficiency of an IC engine is defined as

(A) brake power/indicated power

(B) indicated thermal efficiency/mechanical efficiency

(C) actual (indicated) thermal efficiency/air-standard efficiency

(D) volumetric efficiency × mechanical efficiency

Q14 NAT

An air-standard Diesel cycle has a compression ratio of 18, with air at 300 K at the start of compression. 1200 kJ/kg of heat is added at constant pressure (c_p = 1.005 kJ/kg·K, γ = 1.4). The cut-off ratio (round off to three decimal places) is _____

Q15 MCQ

The compression ratio of a petrol (SI) engine is limited mainly by

(A) the strength of the connecting rod

(B) the volumetric efficiency

(C) the cooling system capacity

(D) detonation (knock) of the end gas

Answer Key

Q	Type	Answer
Q1	MCQ	(B) the compression ratio and γ only
Q2	NAT	7.36 (accept 7.30 to 7.42)
Q3	MCQ	(C) V_3/V_2
Q4	NAT	59.05 (accept 58.70 to 59.40)
Q5	MCQ	(D) CI engines work at much higher compression ratios (about 15–22), as only air is compressed
Q6	NAT	27.43 (accept 27.20 to 27.65)
Q7	MCQ	(A) $(W - S)\pi DN/60\,000$
Q8	NAT	6.29 (accept 6.24 to 6.34)
Q9	MCQ	(A) during 2–3 at constant volume and during 3–4 at constant pressure
Q10	NAT	63.95 (accept 63.60 to 64.30)
Q11	MCQ	(B) the flow losses (friction and choking) across the intake valve rise sharply with the gas velocity
Q12	NAT	7.41 (accept 7.36 to 7.46)
Q13	MCQ	(C) actual (indicated) thermal efficiency/air-standard efficiency
Q14	NAT	2.253 (accept 2.240 to 2.265)
Q15	MCQ	(D) detonation (knock) of the end gas

Detailed Solutions

Q1 **Answer: (B) the compression ratio and γ only**

Concept: $\eta_{Otto} = 1 - 1/r^{\gamma-1}$.

Only the compression ratio r and the ratio of specific heats γ appear: **(B)**.

The heat supplied changes the work and the MEP, not the air-standard efficiency.

Exam tip: Real SI engines cannot raise r beyond about 10–12 because of knock, which caps their efficiency.

Q2 **Answer: 7.36 (accept 7.30 to 7.42)**

Concept: Invert $\eta = 1 - 1/r^{\gamma-1}$.

$r^{0.4} = 1/(1 - 0.55) = 2.222$

$r = 2.222^{2.5} =$ **7.36**

Exam tip: Each step up in r gives a smaller gain in η: going from r = 7.36 to 10 adds only about 5 percentage points.

Q3 Answer: (C) V_3/V_2

Concept: Cut-off ratio ρ is the volume ratio over which fuel is burnt (heat is added) at constant pressure.

ρ = **V_3/V_2 (C)**. V_1/V_2 is the compression ratio r.

The expansion ratio is $V_4/V_3 = r/\rho$.

Exam tip: Cut-off as a percentage of stroke = $(\rho - 1)/(r - 1) \times 100$.

Q4 Answer: 59.05 (accept 58.70 to 59.40)

Concept: $\eta_{Diesel} = 1 - (1/r^{\gamma-1})(\rho^{\gamma} - 1)/[\gamma(\rho - 1)]$.

$16^{0.4} = 3.0314$, so $1/r^{0.4} = 0.3299$

$2.5^{1.4} = 3.6067$; $(3.6067 - 1)/(1.4 \times 1.5) = 1.2413$

$\eta = 1 - 0.3299 \times 1.2413 =$ **59.05 %**

Exam tip: The Otto cycle with r = 16 would give 67.0 %: the bracket term is the Diesel penalty.

Q5 Answer: (D) CI engines work at much higher compression ratios (about 15–22), as only air is compressed

Concept: In a CI engine only air is compressed and fuel is injected at the end, so there is no knock limit on compression.

Compression ratios of 15–22 against 8–12 for SI engines more than make up for the Diesel cycle penalty: **(D)**.

CI engines also run lean and unthrottled, reducing pumping losses at part load.

Exam tip: Compare cycles at equal maximum pressure to see why: the Diesel cycle then wins.

Q6 Answer: 27.43 (accept 27.20 to 27.65)

Concept: $\eta_{bth} = bp/(\dot{m}_f \times CV)$, with $bp = \eta_m \times ip$.

$bp = 0.8 \times 60 = 48$ kW

Fuel energy $= 15 \times 42\,000/3600 = 175$ kW

$\eta_{bth} = 48/175 =$ **27.43 %** (indicated thermal efficiency = 60/175 = 34.29 %)

Exam tip: $\eta_{bth} = \eta_m \times \eta_{ith}$.

Q7 Answer: (A) (W - S)πDN/60 000

Concept: Brake torque T = (W − S) × D/2 (net rope tension × drum radius).

$bp = 2\pi NT/60 = 2\pi N(W - S)(D/2)/60$ W $= (W - S)\pi DN/60$ W

In kW: **(W − S)πDN/60 000 (A)**, with D = drum + rope diameter.

Exam tip: All the brake power is dissipated as heat in the drum, which must be water-cooled.

Q8 Answer: 6.29 (accept 6.24 to 6.34)

Concept: $bp = p_{mb} \times V_s \times (N/2)/60 \times$ number of cylinders for a four-stroke engine.

$V_s = (\pi/4)(0.09)^2 \times 0.1 = 6.362 \times 10^{-4}$ m³ per cylinder

Swept volume rate $= 4 \times 6.362 \times 10^{-4} \times 1500/60 = 0.06362$ m³/s

$p_{mb} = 40/0.06362 = 628.8$ kPa = **6.29 bar**

Exam tip: Do not forget N/2 for four-stroke engines; a two-stroke engine uses N.

Q9 **Answer: (A) during 2–3 at constant volume and during 3–4 at constant pressure**

Concept: The dual cycle burns part of the fuel at constant volume (near TDC) and the rest at constant pressure.

2–3: vertical line, constant-volume heat addition; 3–4: horizontal line, constant-pressure heat addition: **(A)**.

4–5 is isentropic expansion and 5–1 constant-volume heat rejection.

Exam tip: With pressure ratio $\alpha = 1$ it becomes the Diesel cycle; with cut-off ratio $\rho = 1$ it becomes the Otto cycle.

Q10 **Answer: 63.95 (accept 63.60 to 64.30)**

Concept: $\eta = 1 - (1/r^{\gamma-1})(\alpha\rho^{\gamma} - 1)/[(\alpha - 1) + \gamma\alpha(\rho - 1)]$.

$15^{0.4} = 2.9542$, so $1/r^{0.4} = 0.3385$

$\alpha\rho^{\gamma} - 1 = 1.4 \times 1.7641 - 1 = 1.4697$; denominator $= 0.4 + 1.4 \times 1.4 \times 0.5 = 1.38$

$\eta = 1 - 0.3385 \times 1.0650 =$ **63.95 %**

Exam tip: Check against limits: Otto ($r = 15$) gives 66.15 %, so the dual value must be lower.

Q11 **Answer: (B) the flow losses (friction and choking) across the intake valve rise sharply with the gas velocity**

Concept: Volumetric efficiency = actual air inducted/(swept volume × ambient density).

At high speed the gas velocity through the valve rises, the pressure drop grows roughly with velocity squared and the flow can choke: **(B)**.

At low speed, back-flow through the late-closing intake valve lowers it instead, so η_v peaks at a mid speed.

Exam tip: Valve timing (inlet closing after BDC) uses the inertia of the charge to raise η_v.

Q12 **Answer: 7.41 (accept 7.36 to 7.46)**

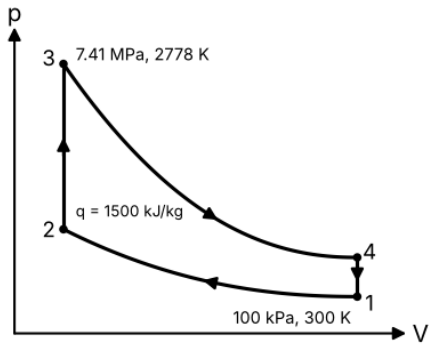
Concept: Isentropic compression, then constant-volume heat addition; use $p_3v_3/T_3 = p_1v_1/T_1$ with $v_3 = v_1/r$.

$T_2 = 300 \times 8^{0.4} = 689.2$ K

$T_3 = 689.2 + 1500/0.718 = 2778.4$ K

$p_3 = p_1\, r\, T_3/T_1 = 100 \times 8 \times 2778.4/300 = 7409$ kPa = **7.41 MPa**

Exam tip: $p_2 = 100 \times 8^{1.4} = 1838$ kPa; heat addition multiplies it by $T_3/T_2 = 4.03$.



Q13 **Answer: (C) actual (indicated) thermal efficiency/air-standard efficiency**

Concept: Relative (efficiency) ratio compares the real engine with its ideal air-standard cycle at the same compression ratio.

$\eta_{rel} = \eta_{actual}/\eta_{air\text{-}standard}$: **(C)**. Typical values are 0.6–0.8.

Option (A) is the mechanical efficiency.

Exam tip: If η_{ith} = 30 % and the Otto value is 56.5 %, η_{rel} = 0.53.

Q14 **Answer: 2.253 (accept 2.240 to 2.265)**

Concept: In constant-pressure heat addition, $\rho = V_3/V_2 = T_3/T_2$.

$T_2 = 300 \times 18^{0.4} = 953.3$ K

$T_3 = 953.3 + 1200/1.005 = 2147.3$ K

$\rho = 2147.3/953.3 =$ **2.253**

Exam tip: More heat (higher load) means a larger cut-off ratio and a lower Diesel-cycle efficiency.

Q15 **Answer: (D) detonation (knock) of the end gas**

Concept: In an SI engine the fuel–air mixture is compressed together.

At high compression ratios the end gas self-ignites before the flame reaches it, causing **knock (D)**.

Higher-octane fuel, a compact chamber and cooler intake allow higher r.

Exam tip: In CI engines the opposite holds: a higher compression ratio reduces diesel knock by shortening the ignition delay.

What next?

Day 49 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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Next, Day 50/90: Applied Thermodynamics, Refrigeration & Air Conditioning. Comment **GATE90** on our reel to get all links.
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